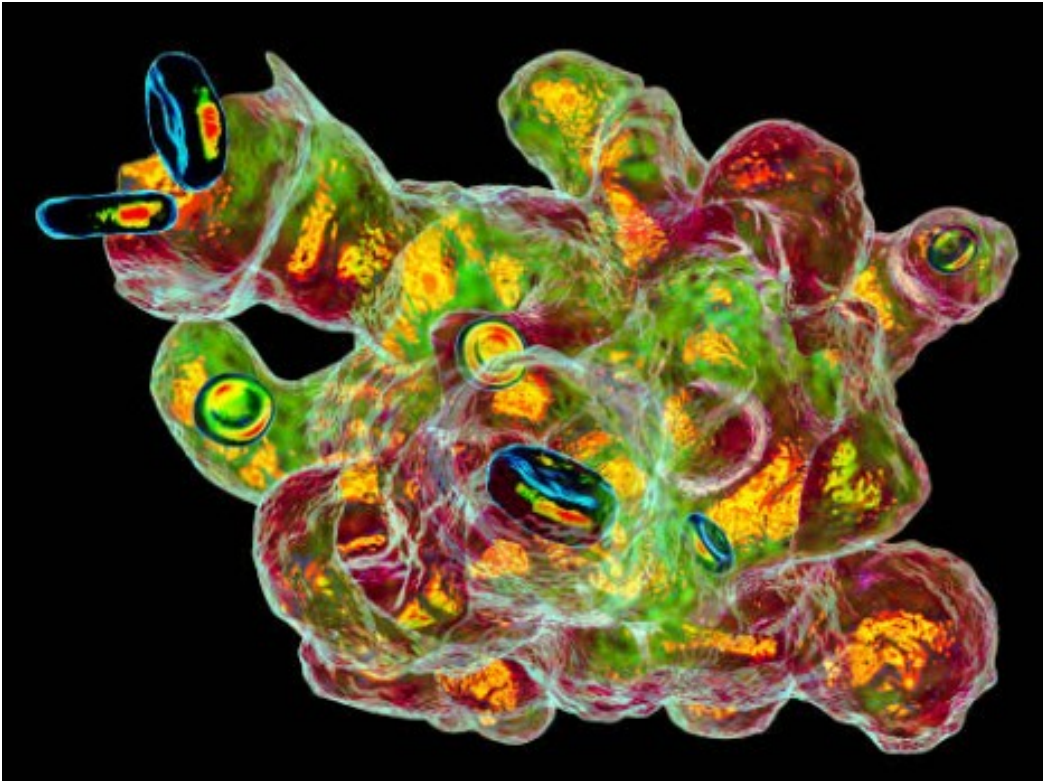


Amoebiasis



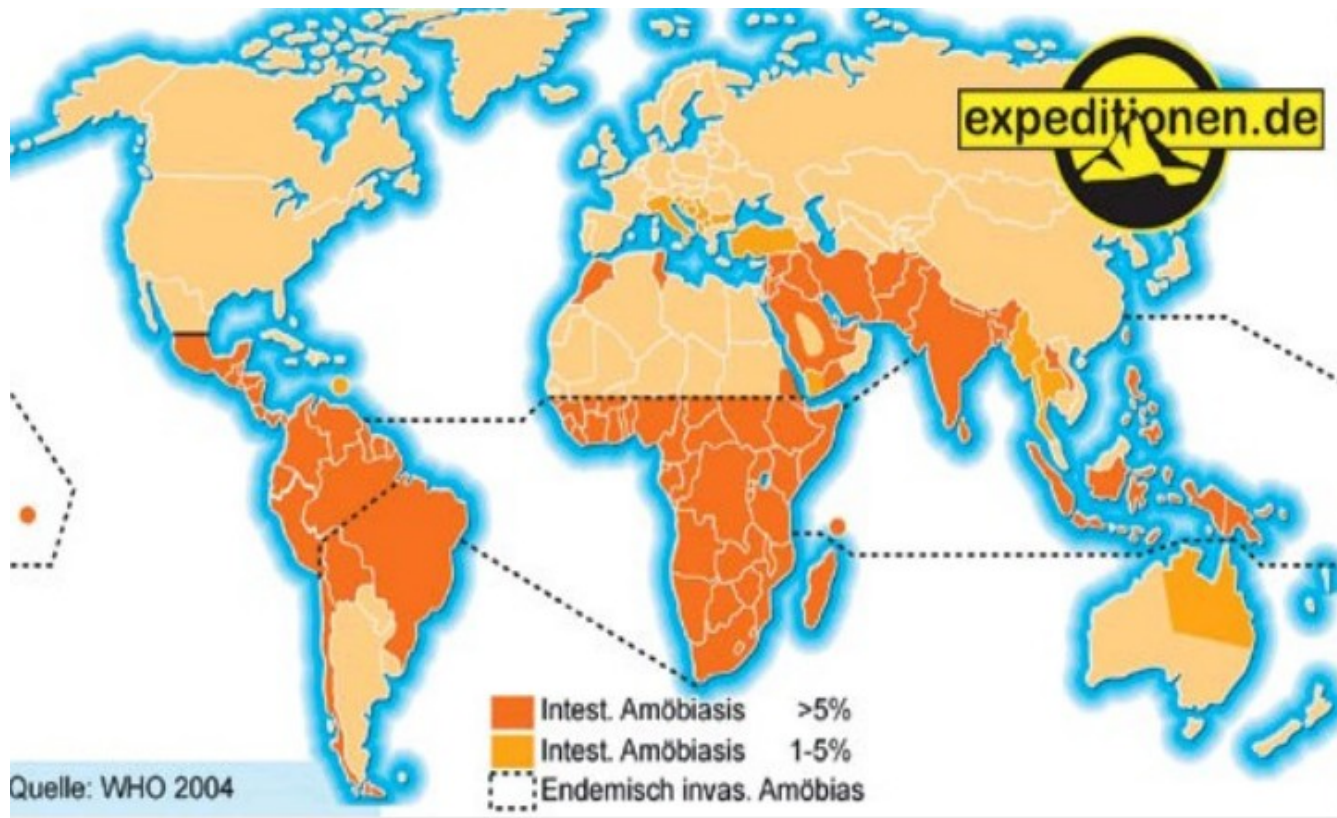
Dr. Yehenew A.

DMIP

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Entamoeba histolytica

- The most pathogenic among other intestine amoeba spp and cause Amebiasis

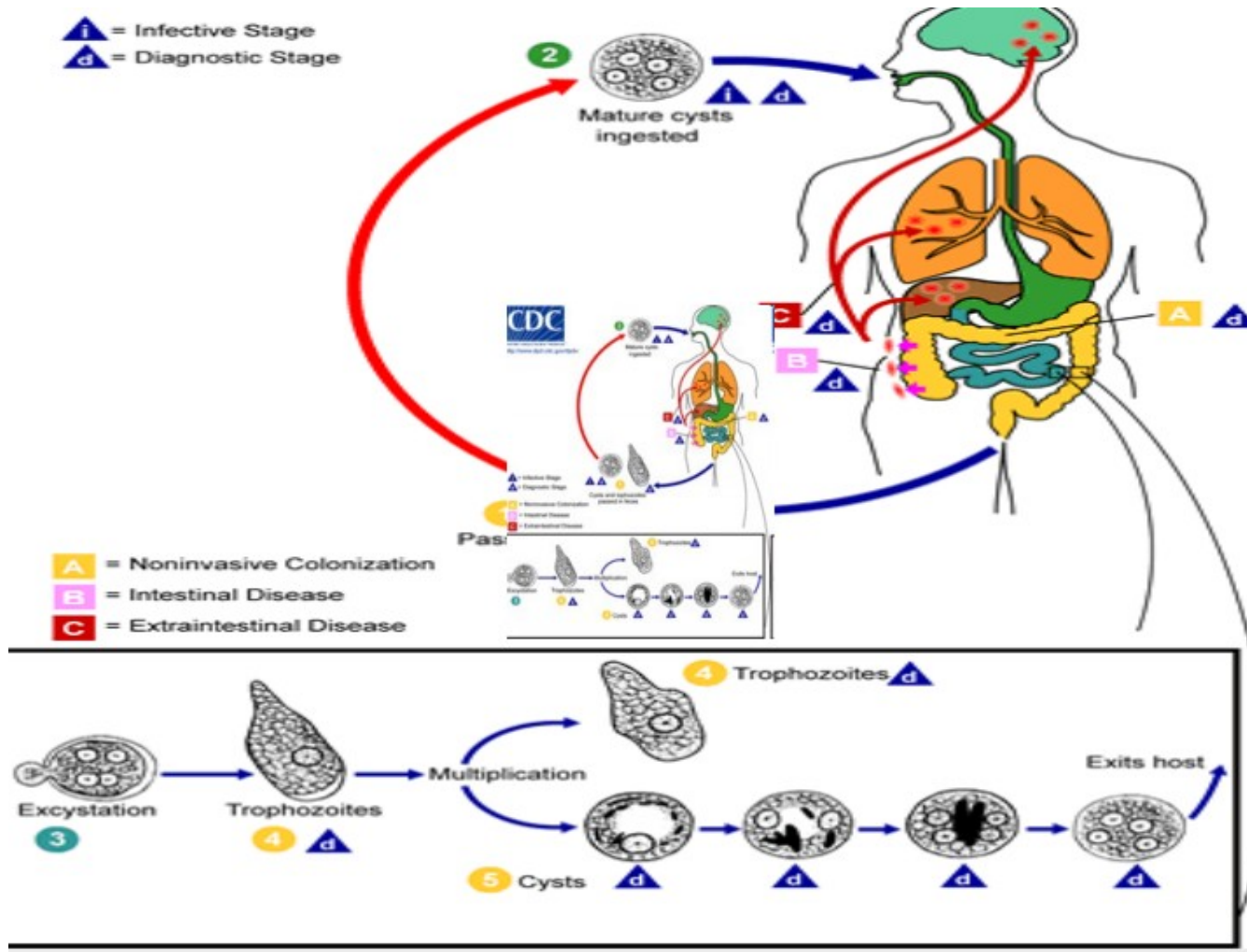


Transmission



- Cyst ingestion
 - Food
 - Water
 - Poor personal hygiene
 - E t c

E. histolytica Life cycle



- Cyst ex-cyst at the small intestine into the trophozoite
- Migrate to the large intestine
 - Binary fusion- multiplication
 - Both stages (cyst and trophozoite) excrete through feaces

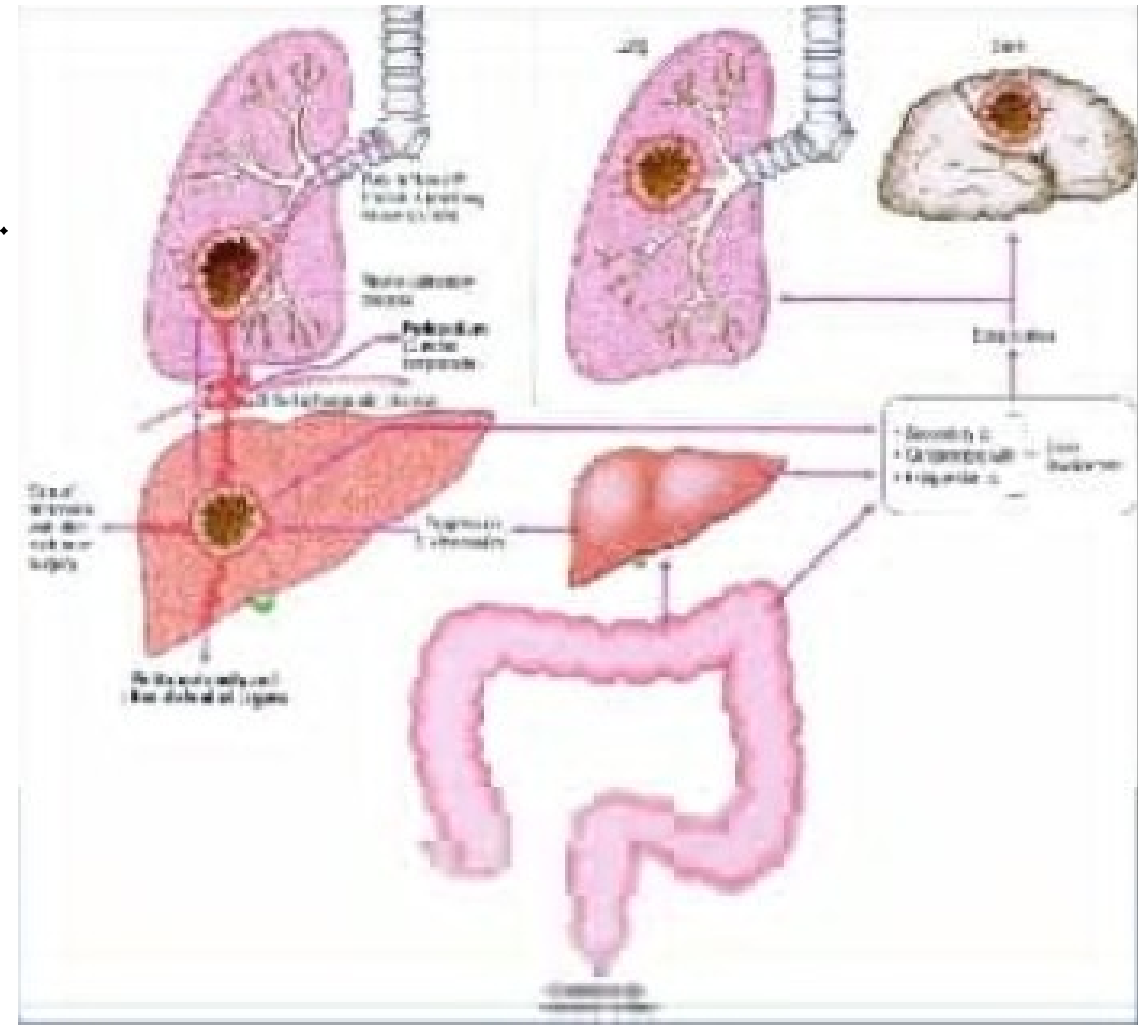
Pathogenesis and Clinical manifestation

1. Primary or intestinal Lesions

- The infection is limited entirely to the large intestine, the initial site of location of the parasite.

2. Secondary or Metastatic Lesions—

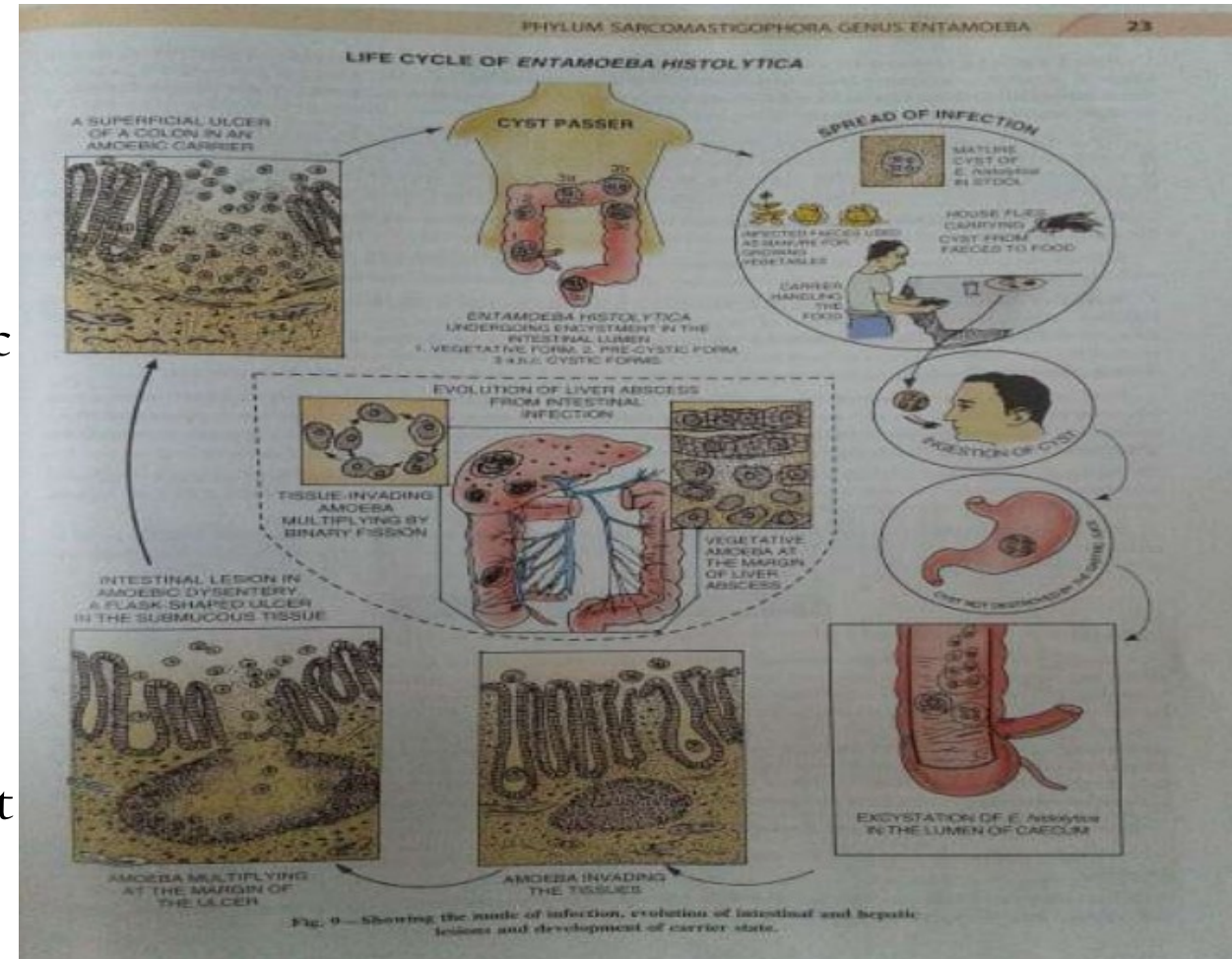
- The extra colonic areas where the trophozoites of *E.histolytica*
 - Can migrate and produce lesions include
 - Liver,
 - Lungs, and
 - Brain



Pathogenesis and Clinical manifestation

I. Intestinal amoebiasis

- Cause disease only when they **invade the intestinal** tissues.
 - About 10% of cases of infection
- **Differentiation** between pathogenic and non pathogenic strains
- The **metacystic trophozoites** **penetrate** the columnar epithelial cells
 - Facilitated by the motility and **tissue enzyme histolysin**
- **Mucosal penetration** causes amoebic ulcer confined to the colon, being most numerous in the **caecum** and next in the **sigmoido-rectal** region.
- The abscess breakdown to form an ulcer

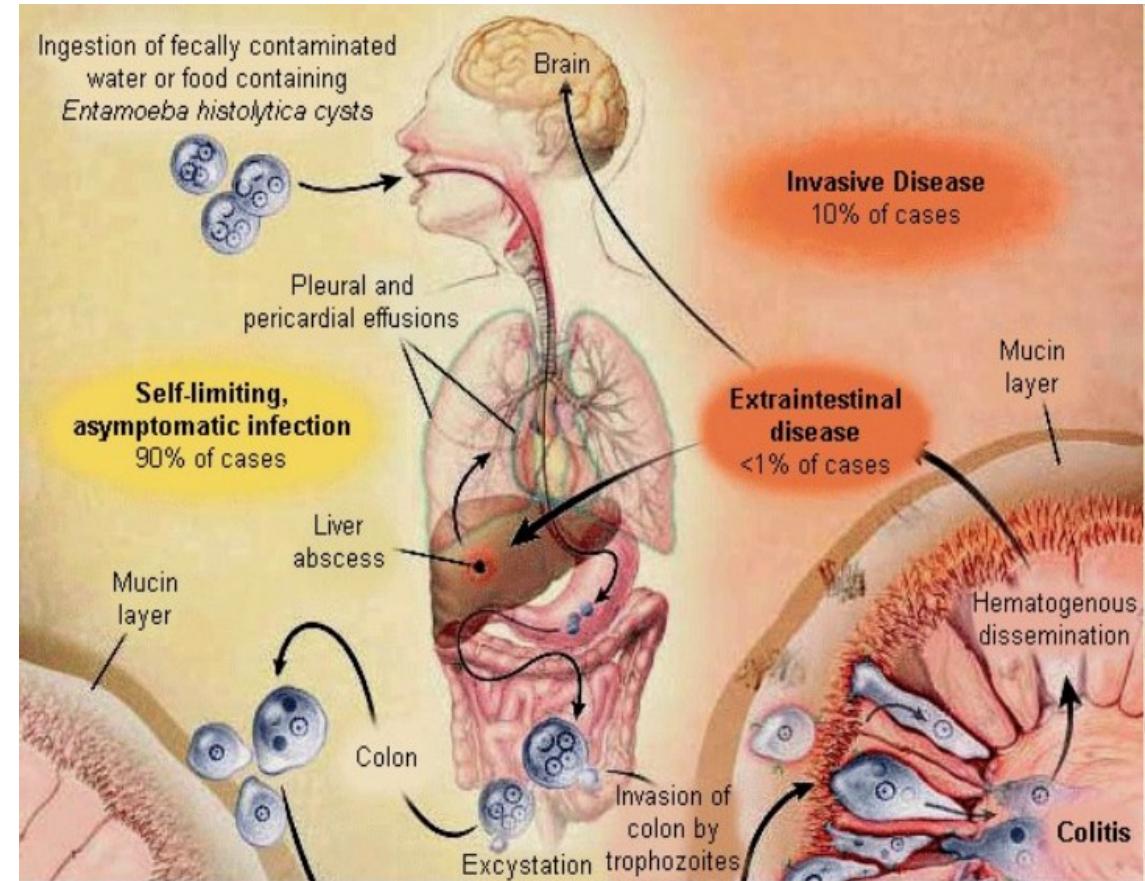


Pathogenesis and Clinical manifestation

2. Extra-intestinal Amoebiasis

2.1. Hepatic amoebiasis/ amoebic hepatitis

- In few dysenteric cases- **trophozoites reach the liver**
- About **5-10%** of persons with intestinal amoebiasis
 - **Develop liver abscesses** (impairment of the liver function without detectable)
- Amoebic hepatitis may be due to **repeated invasion** by amoebae from **active colonic infection/ toxic substances**
 - This is not directly by the amoebae but by the lysosomal enzymes and cytokines from the inflammatory cells surrounding the trophozoites

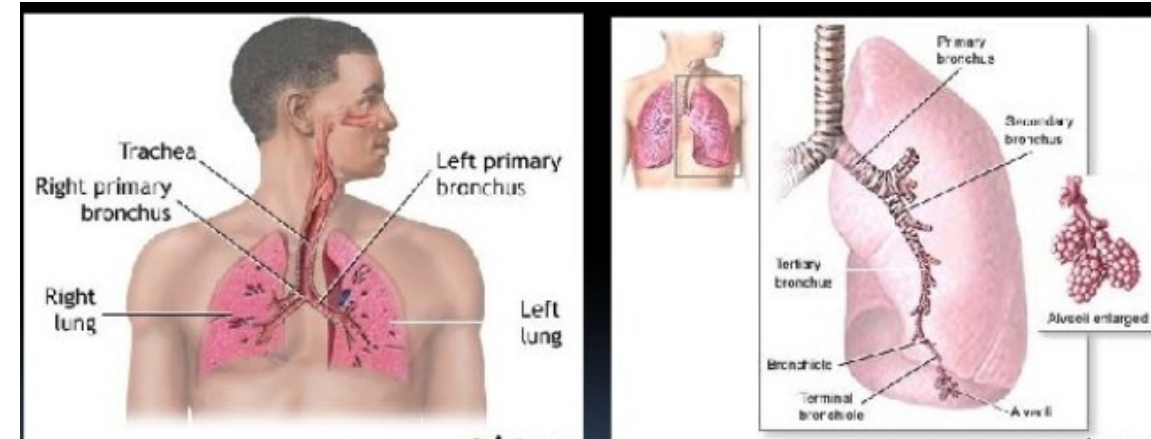


- Untreated liver abscesses tend to rupture into the adjacent tissues
 - Into the lung or pleural cavity, Pericardium, peritoneal cavity,
 - Stomach and skin

Pathogenesis and Clinical manifestation

2.2. Pulmonary Amoebiasis

- Very rarely primary amoebiasis of the lung may occur by **direct hematogenous spread from the colon** bypassing the liver
 - The lower part of the **right lung** is the usual area affected.
 - In Hepatobronchial fistula: Chocolate Brown Sputum
 - Leads to Pleuritis chest pain, Dyspnea, Non productive cough



2.3. Cutaneous Amoebiasis

- It occurs by direct extension around anus
- Discharging sinuses from amoebic abscesses

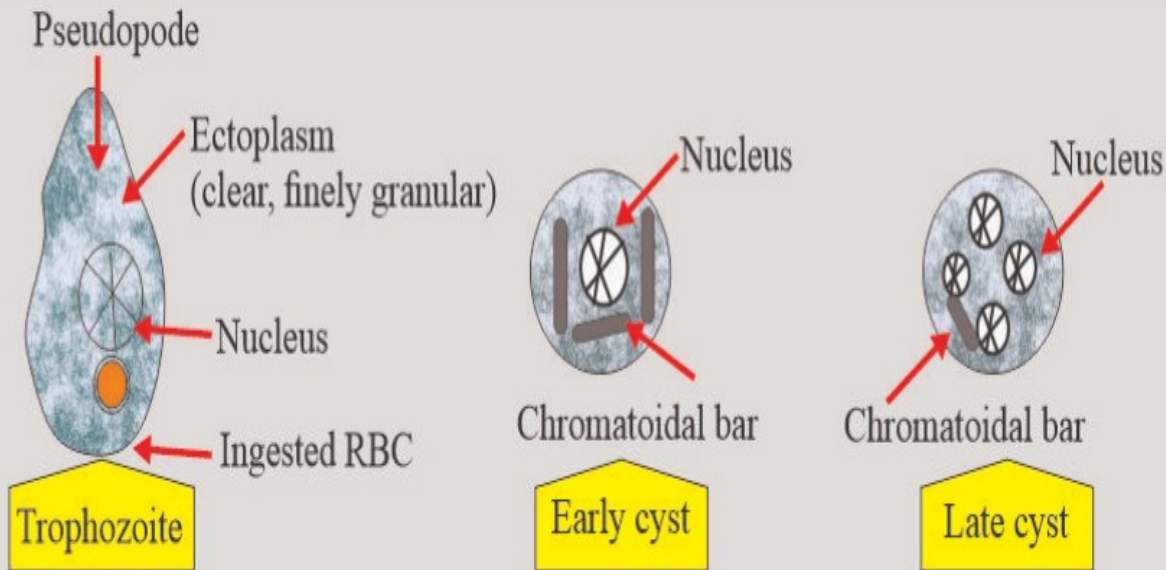
2.4. Metastatic Amoebiasis

- Spread to distant organs through blood and lymph
- When brain is affected its fatal : **Cerebral Amoebiasis**



Radiographic and Pathological Features of Extraintestinal Amebiasis.

Diagnosis



1. Motile
2. 15 to 60 μm
average 25 μm
3. Single pseudopodium
4. Mitosis (Binary fission)
in the host gut

1. An average size is 12 μm
2. With single nucleus
3. Chromatoidal bars
(deposit of nucleic acid)

1. Nuclei 4 vesicular
2. Infective form
3. Resistant to environment
4. Can survive 3 months
5. Thermal death point is 50 C

Table 1. Sensitivity of Tests for the Diagnosis of Amebiasis.*

Test	Colitis	Liver Abscess
		percent
Microscopy		
Stool	25–60	10–40
Abscess fluid	NA	≤ 20
Antigen detection		
Stool	90	~40
Serum	65 (early)	~100 (before treatment)
Abscess fluid	NA	~40
Indirect hemagglutination (antibody)		
Serum obtained during acute illness	70	70–80
Serum obtained during convalescence	>90	>90

Treatment

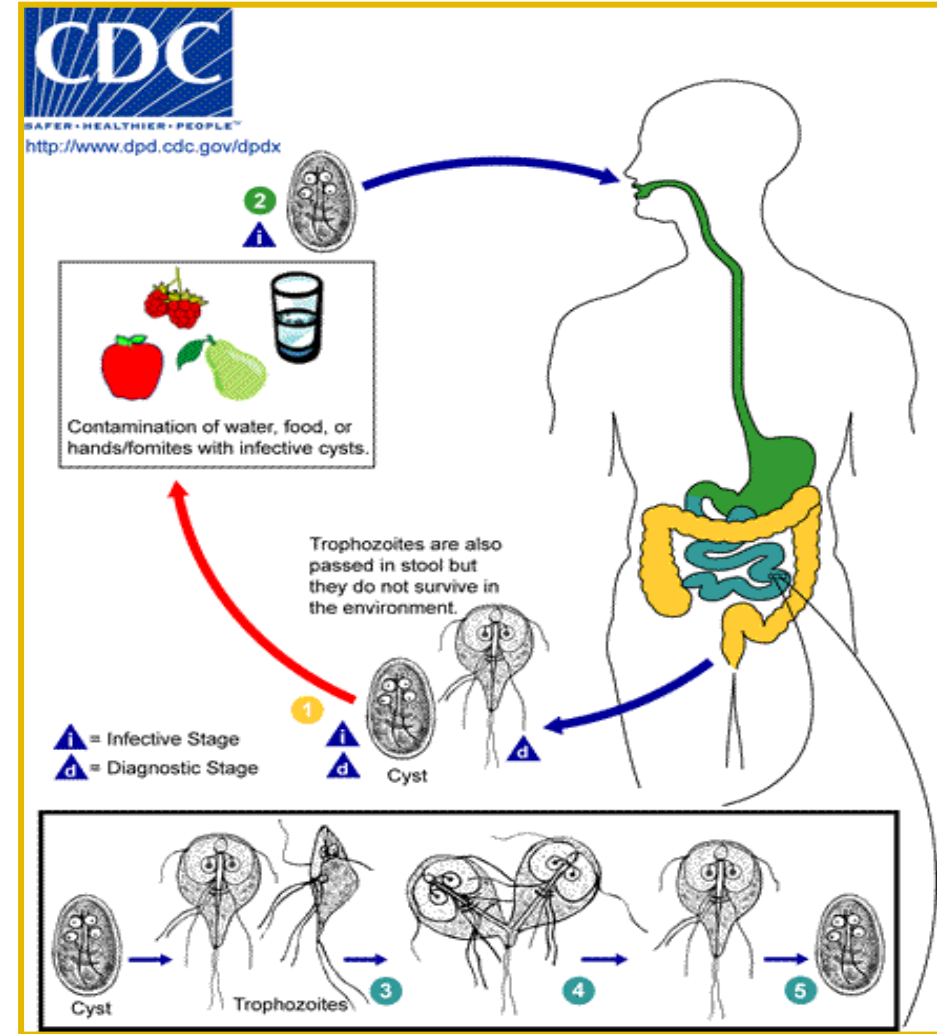
Table 2. Drug Therapy for the Treatment of Amebiasis.*

Drug	Adult Dosage	Pediatric Dosage	Side Effects
Amebic liver abscess			
	give		
Metronidazole	750 mg orally 3 times a day for 7–10 days	35–50 mg/kg of body weight/day in 3 divided doses for 7–10 days	Primarily gastrointestinal: anorexia, nausea, vomiting, diarrhea, abdominal discomfort, or unpleasant metallic taste; disulfuram-like intolerance reaction with alcohol; rarely, neurotoxicity, including seizures, peripheral neuropathy, dizziness, confusion, irritability
	or		
Tinidazole†	800 mg orally 3 times a day for 5 days	60 mg/kg/day (maximum, 2 g) for 5 days	Primarily gastrointestinal and disulfuram-like intolerance reaction as for metronidazole
	followed by a luminal agent		
Paromomycin	25–35 mg/kg/day in 3 divided doses for 7 days	25–35 mg/kg/day in 3 divided doses for 7 days	Primarily gastrointestinal: diarrhea, gastrointestinal upset
	or second-line agent		
Diloxanide furoate‡	500 mg orally 3 times a day for 10 days	20 mg/kg/day in 3 divided doses for 10 days	Primarily gastrointestinal: flatulence, nausea, vomiting, pruritus, urticaria
Amebic colitis			
	give		
Metronidazole	750 mg orally 3 times a day for 7–10 days	35–50 mg/kg/day in 3 divided doses for 7–10 days	As for amebic liver abscess
	followed by a luminal agent (as for amebic liver abscess)		
Asymptomatic intestinal colonization			
	give		
Paromomycin	25–35 mg/kg/day in 3 divided doses for 7 days	25–35 mg/kg/day in 3 divided doses for 7 days	Primarily gastrointestinal: diarrhea, gastrointestinal upset
	or second-line agent		
Diloxanide furoate‡	500 mg orally 3 times a day for 10 days	20 mg/kg/day in 3 divided doses for 10 days	Primarily gastrointestinal: flatulence, nausea, vomiting, pruritus, urticaria

Giardiasis (G.lamblia)

Giardiasis (*G. lamblia*)

- *Global but high in tropical countries*
- Transmission
 - Fecal-oral transmission
 - Ingestion of contaminated water
- Significant pathogen for people with:
 - Malnutrition
 - Immunodeficiencies
 - Cystic fibrosis



Giardiasis

Gastrointestinal manifestations

○ Abrupt onset (rare)

- a small number of persons develop abrupt onset of :

• Explosive watery diarrhea	• Vomiting
• Abdominal cramps	• Fever
• Foul flatus	• Malaise

- these symptoms last 3-4 days

Giardiasis: Gastrointestinal manifestation

○ Subacute syndrome (More common)

- After the symptoms of abrupt onset
- Most patients experience a more insidious onset of symptoms, which are recurrent or resistant.
- Stools become malodorous, mushy, and greasy
- Watery diarrhea may alternate with soft stools or even constipation

Gastrointestinal manifestation

○ Upper GI symptoms:

- often exacerbated by eating
- accompany stool changes
- may be present in the absence of soft stools.

○ These include:

- upper and midabdominal cramping
- nausea
- early satiety
- bloating
- substernal burning
- acid indigestion.

Extraintestinal manifestations

- Are rare
- Include allergic manifestations such as:
 - Urticaria
 - Erythema multiforme
 - Bronchospasm
 - Reactive arthritis
 - Biliary tract disease
- The etiology is likely a result of :
 - Host immune system activation
 - Cross-reactivity/molecular mimicry.

Symptoms of chronic infection

Symptoms of chronic infection

- | | |
|------------|---|
| • Malaise | • Weight loss
(~66% of symptomatic patients) |
| • Nausea | • Post infection lactase deficiency
(2-40% of cases) |
| • Anorexia | • Chronic diarrhea |

Complications of giardiasis

May include the following:

- Development of chronic illness with weight loss
- Malabsorption syndrome in adults
- Failure to thrive in children
- Disaccharidase deficiency
- Zinc deficiency in schoolchildren ¹
- Growth retardation ²
- Persistent gastrointestinal symptoms ³

The diagnosis

○ Stool examination

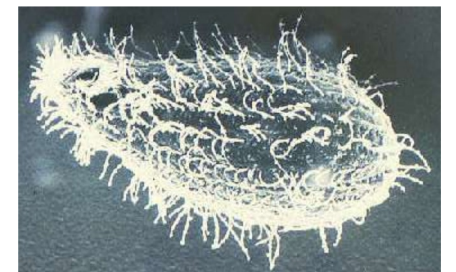
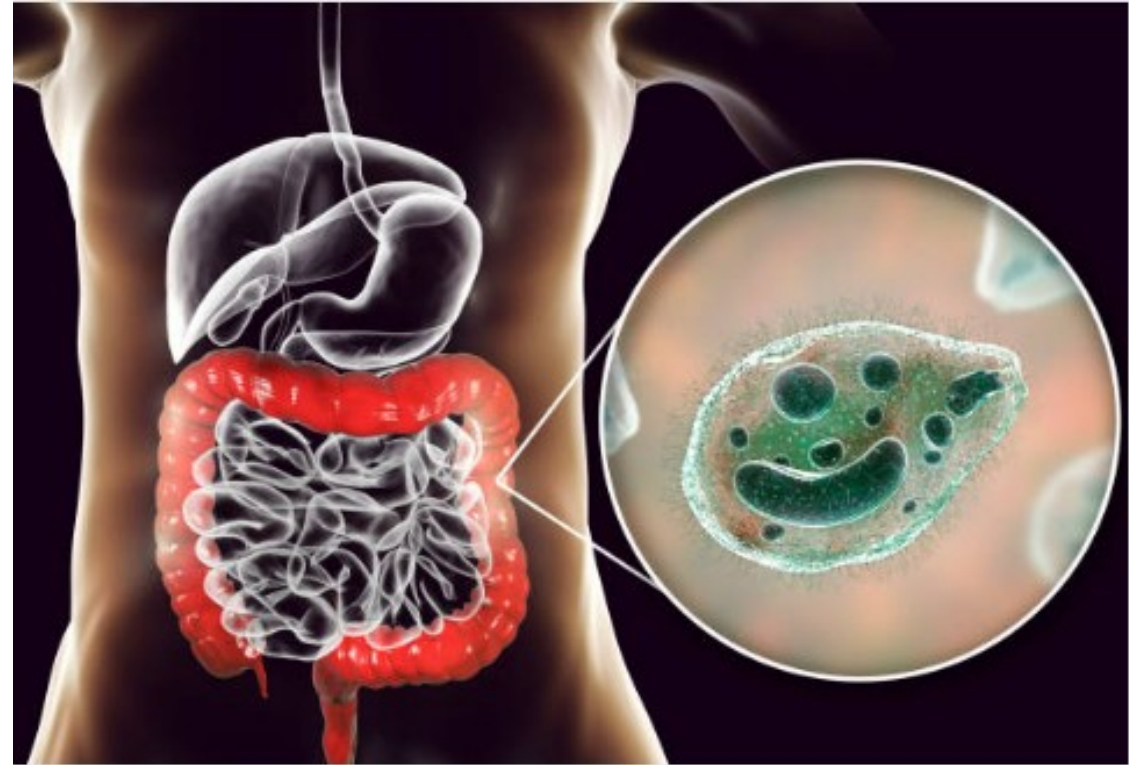
- Ideally, 3 specimens from different days should be examined because of potential variations in fecal excretion of cysts.
- *G intestinalis* is identified in 50–70% of patients after a single stool examination and in more than 90% after 3 stool examinations.

○ Stool ova and parasite (O&P) examination

aids in the diagnosis of giardiasis in 80–85% of patients

Rx: Metronidazole, Tinidazole

Balantidiasis- *Balantidium coli*

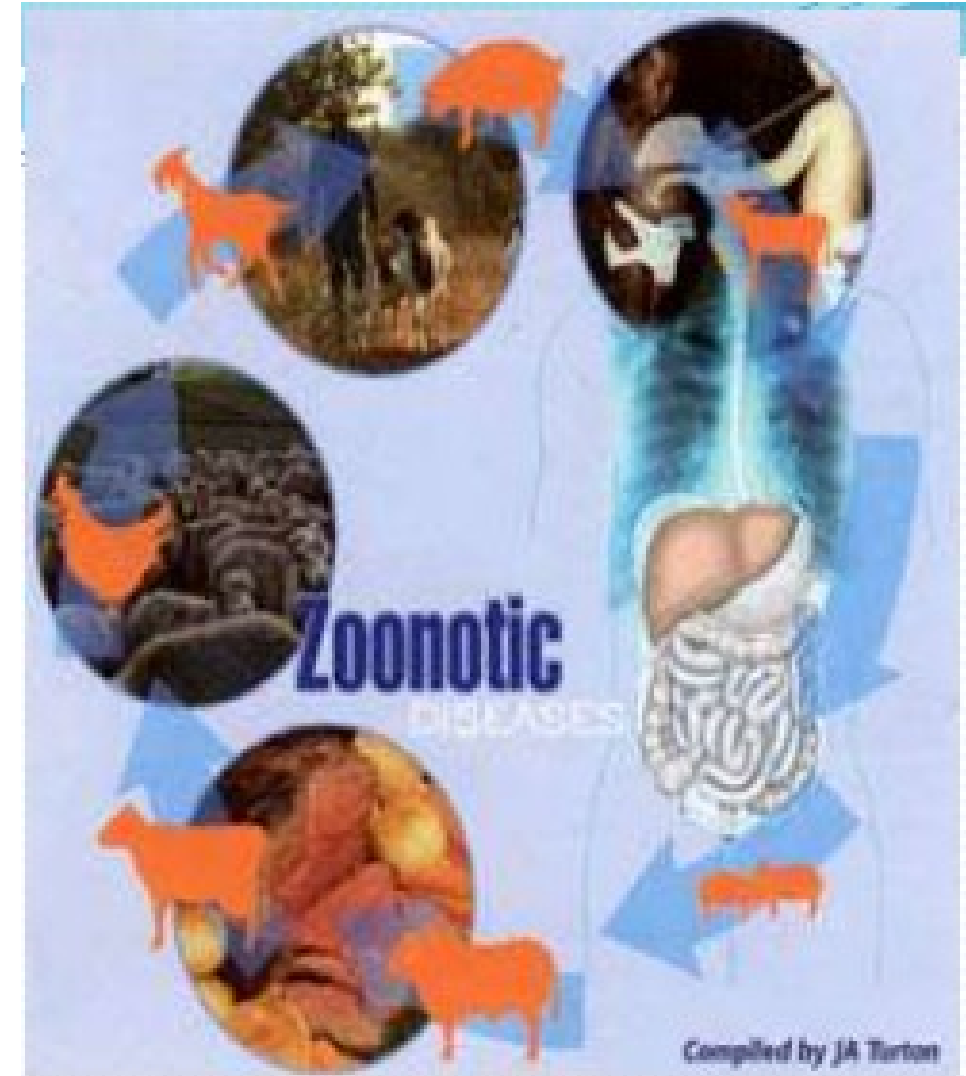


Balantidiosis

- Is a zoonotic disease.
- It is acquired by humans
 - Via the **fecal-oral** route from the normal host, the pig, where it is Asymptomatic.

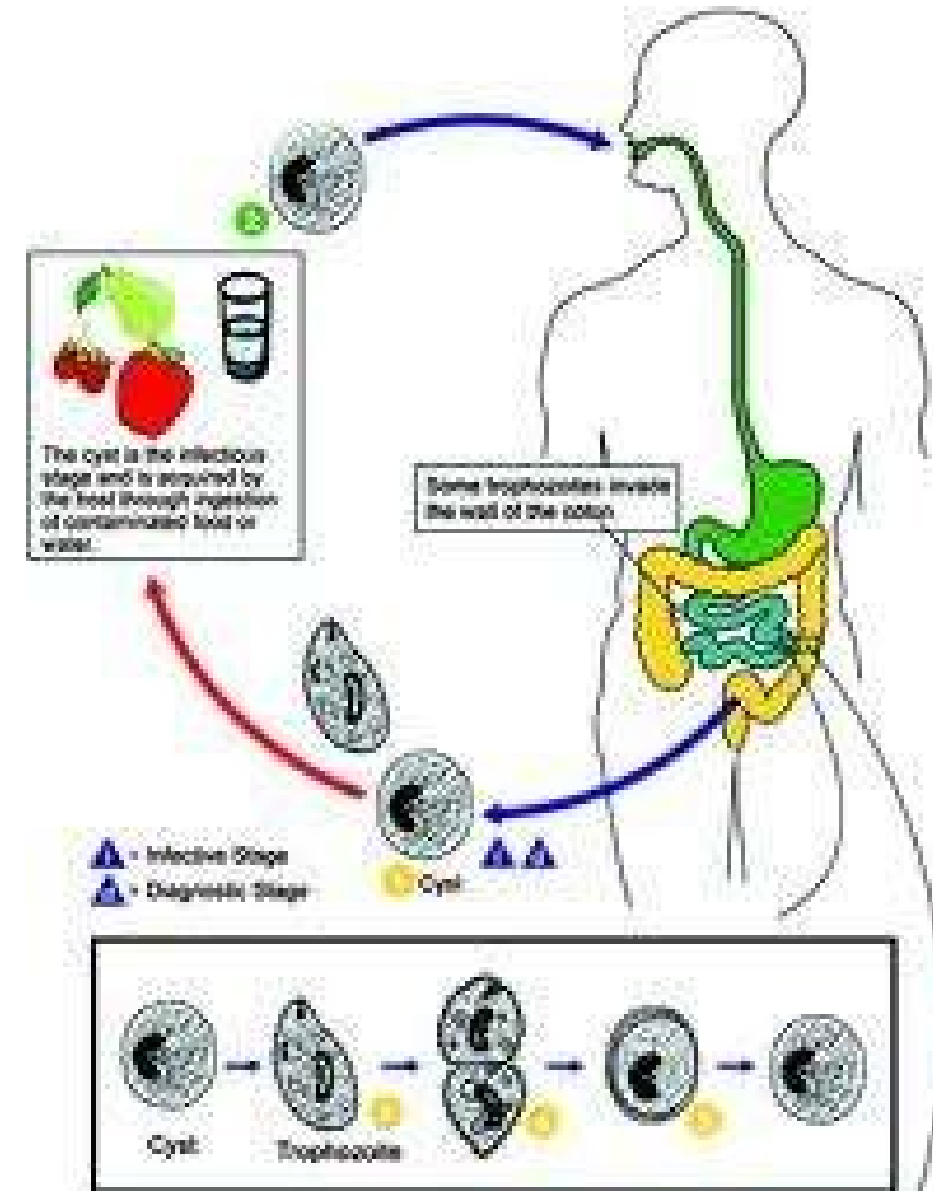


Water is the vehicle for most cases of Balantidiosis



Balantidiasis

- The cyst is the infective stage
- Once the cyst is ingested via water/ food
- There, excystation takes place in small intestine.
- Excystation produces a trophozoite from the cyst stage.



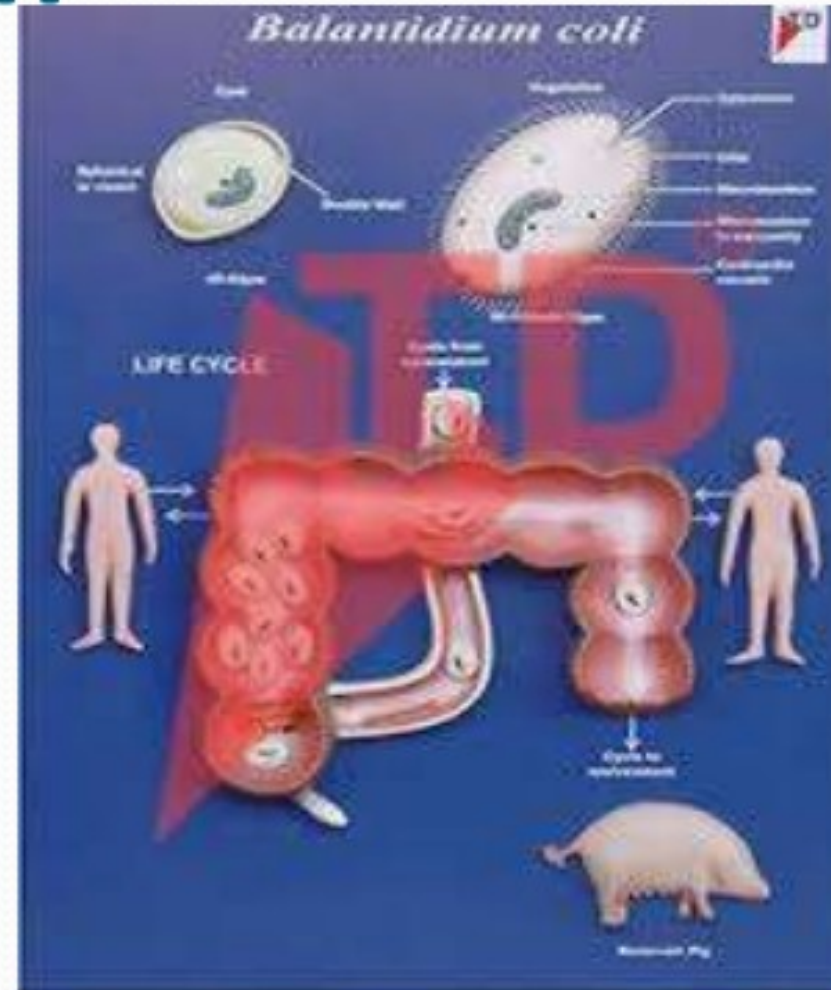
Balantidiasis

Site of Infection

Following are the site of Infection :-

1) Caecum

2) Large Intestine



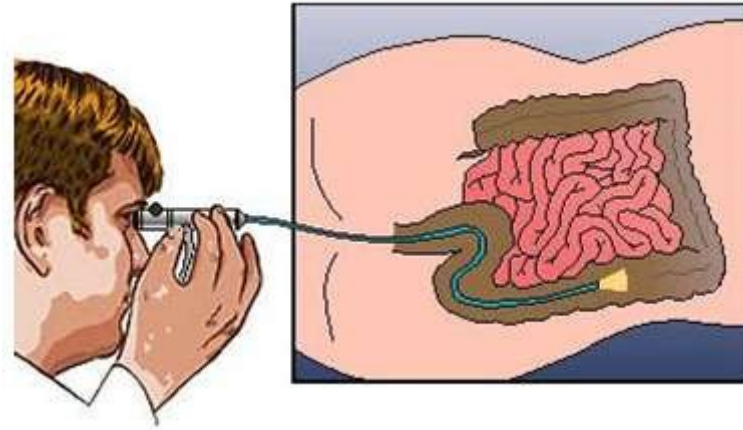
Balantidiasis: Clinical manifestation

- Acute ,even hemorrhagic
- Diarrhea
- Ulceration to gut wall
- Dysentery
- Colitis
- Abdominal pain



Balantidiasis: Diagnosis

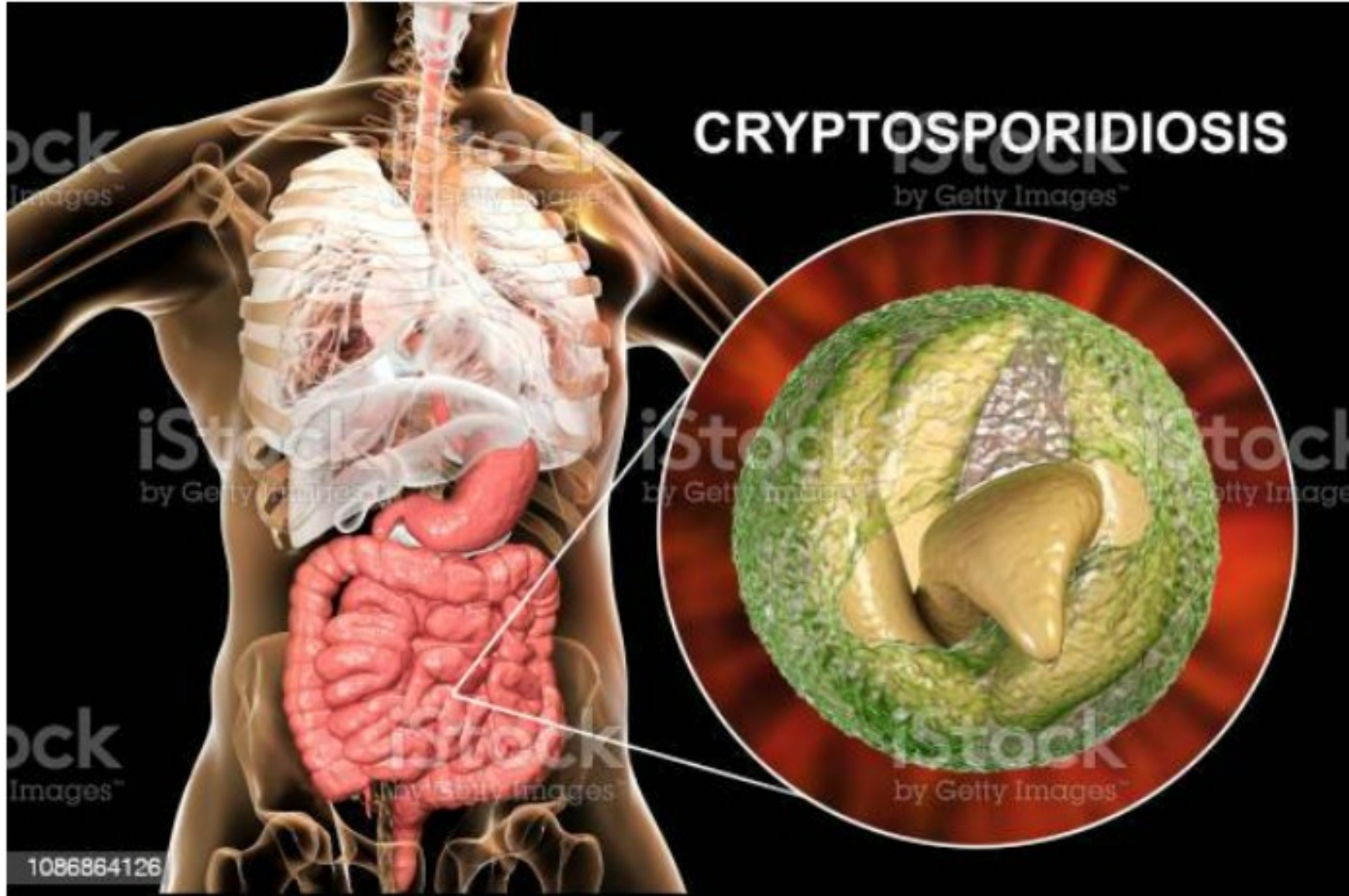
- Examination of patient's stool
- Biopsy
- **Sigmoidoscopy**- is used to visually inspect the last sections of the large intestine



Treatment

- 1) Tetracycline
- 2) Metronidazole
- 3) Iodoquinol

Cryptosporidium parvum



C. parvum

A coccidian parasite occupied in **intestinal tract**

- Disease primarily among **immuno compromised patient**
- Profuse, watery diarrhea associated with AIDS (life threatening)
- But **self-limiting** diarrhea in **immunocompetent** persons

Cryptosporidium..cont

Geog. Dist.: **Worldwide distribution**

Transmission:

- Mainly through the **ingestion sporulated oocysts**
 - Mainly **through contaminated water** & occasionally **food sources**, such as chicken, salad
 - Rarely through **inhalation** of sporulated oocysts
 - **Fecal-oral transmission** (autoinfection called **monoxenous**)
 - Anthroponotic, autoinfection, zoonotic transmission

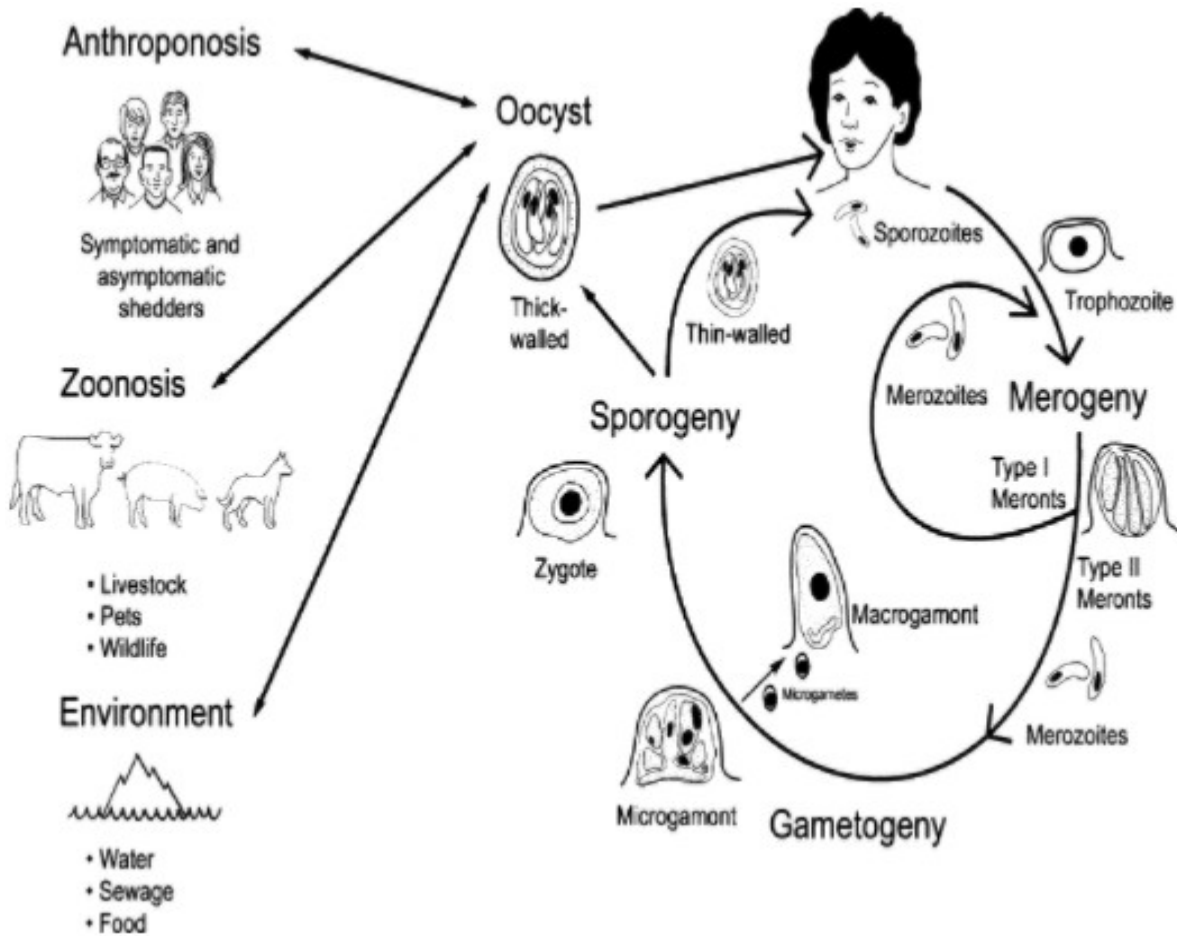
Waterborne Cryptosporidiosis

- Human infection usually **waterborne** and acquired by **fecal-oral route**
- Highest prevalence in areas with **unreliable water and food sanitation**

Factors favoring waterborne cryptosporidiosis

- **Monoxenous** development (restricted to a single host species- **autoinfection**)
- Close associations between **human and animal hosts**
- Large number of oocysts excreted (**100 billion per calf /day**)
- **Low infective dose** (<30)
- Robust oocysts: **resistant to chlorine**

Life Cycle



In the life cycle of *Cryptosporidium*

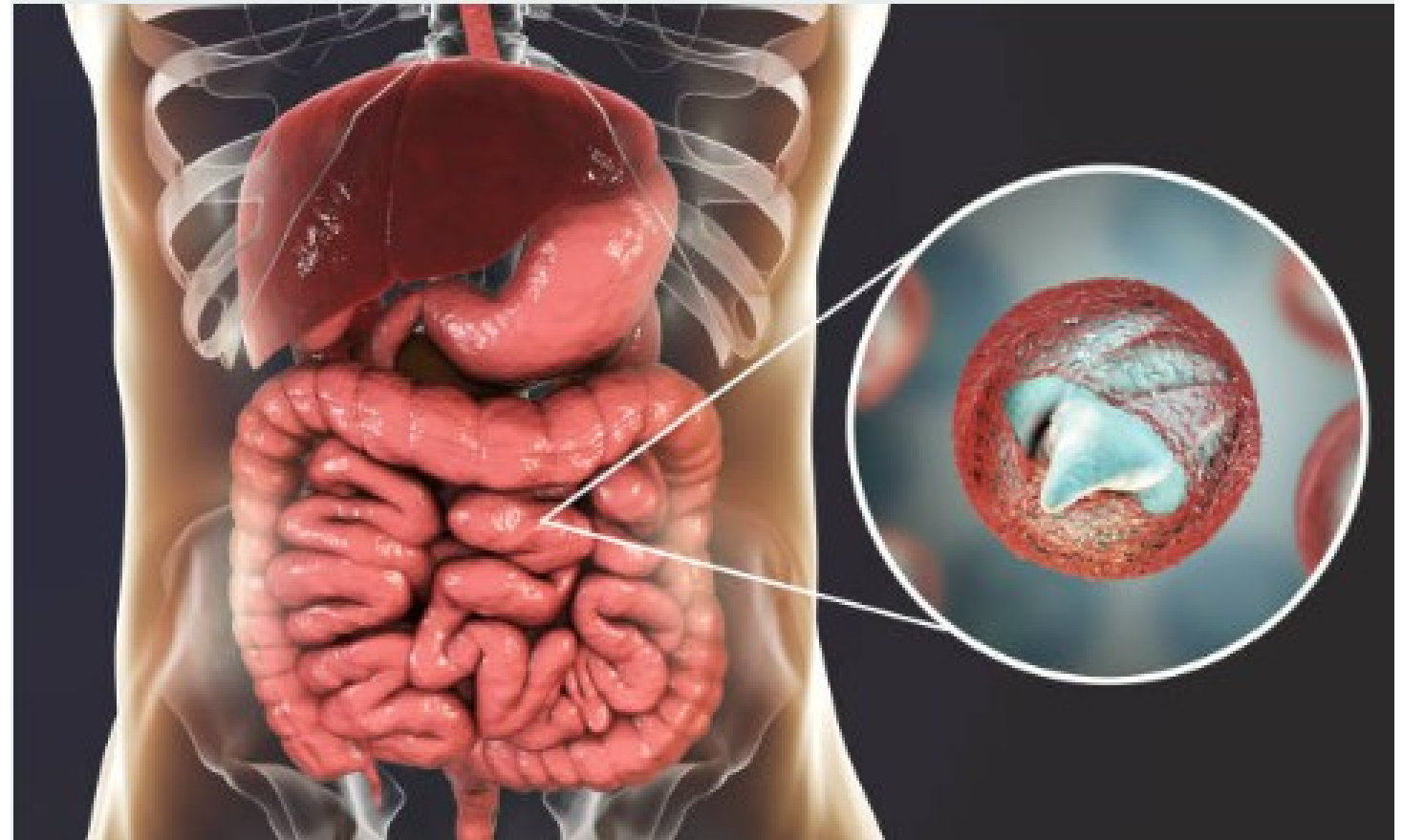
- ❖ Merogony (schizogony): produce merozoites from trophozoites
- ❖ Gametogony (gamogony or gametocytogenesis: produce micro- and macro gametes from merozoite
- ❖ Sporogony: produce sporozoites from gametes
- Thin oocyst (autoinfection) or thick walled oocyst transmission

Disease: Cryptosporidiosis

▪ Extra intestinal infection of the respiratory tract, biliary tract and pancreas may occur.

Pathogenesis

- Enterocytes damaged or killed
 - Villus atrophy (blunting)
 - Enterocyte malfunction
 - Impaired absorption
- Crypt cell hyperplasia
- Inflammation in lamina Propria
 - Impaired absorption
 - Promote secretion



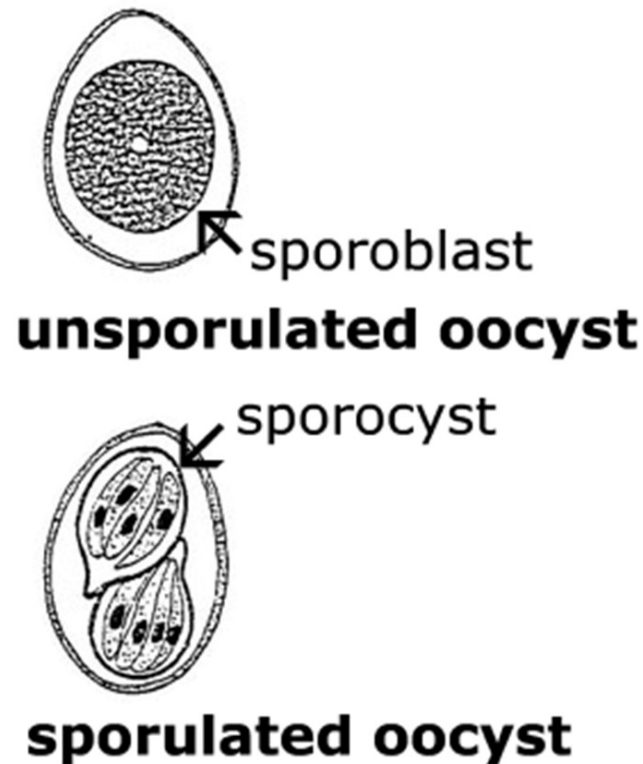
Clinical Features

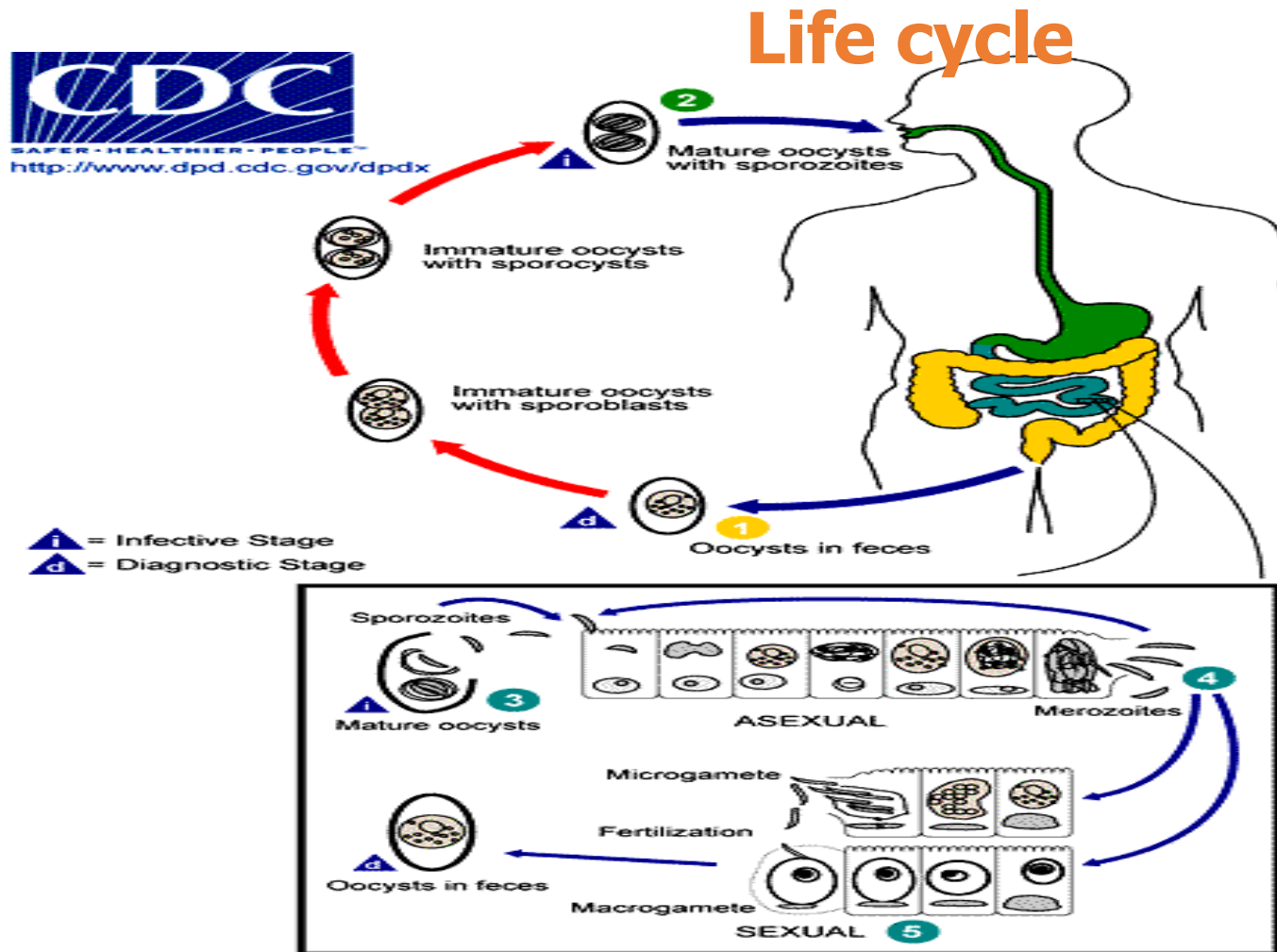
- Incubation period ~ 7 days (2 to 10 days).
- In immunocompetent persons, symptoms are usually short lived (1 to 2 weeks)
- Vary from asymptomatic to severe and life-threatening illness
- Infection usually self-limiting diarrhea of 1-2 weeks duration
 - Characterized by plentiful watery diarrhea, vomiting, intense abdominal pain, anorexia and weakness
- Among immunocompromised patient AIDS (CD4 counts <200/ μ l)
 - Develops severe chronic diarrhea which may last for months even years
 - And cause malabsorption
 - Small intestine: most commonly affected : the lungs, and possibly conjunctiva may be affected.

Isospora belli

Epidemiology:

- Wide geographical distribution (higher prevalence in warmer climates)
- The least common of the three intestinal coccidia that infect humans
- Transmitted by faecally contaminated food and drink with oocyst





Life cycle similar to *Cryptosporidium* except that oocysts mature in environment

Clinical feature

- **Habitat:** intestinal tract probably in the ileum and cecum
- **Disease:** Isosporosis/Intestinal coccidiosis
- Often asymptomatic (seldom reported)
- Symptoms range from mild gastro-intestinal distress to severe dysentery
- Mild self-limiting infection characterized by
 - Fever, colicky abdominal pain, severe diarrhea, steatorrhea (fatty stool) and weight loss.
- Often self-limiting, but can become chronic (wasting, anorexia)
- Symptoms more severe in AIDS patients

Laboratory Diagnosis of Intestinal coccidia

- Microscopy
- Immunological assay
- Molecular methods
- **Safety**
 - *C. parvum* oocysts in stool remain infective for extended periods,
 - Thus specimens should be preserved in 10% buffered formalin to render oocysts nonviable (18 to 24 hrs)

Treatment, prevention and control of intestinal Coccidia

- Paromomycin for *Cryptosporidium*
- Trimethoprim-sulfamethoxazole for *Cyclospora* and *Isospora*
- ❖ Treatment of the case
- ❖ Cooking food items thoroughly
- ❖ Drinking boiled water
- ❖ Limiting the consumption raw vegetables and salads
- ❖ Personal and environmental sanitation
- ❖ Avoiding contact with animals

Thank you